PREPVECTOR
Cognitive Assessment Platform

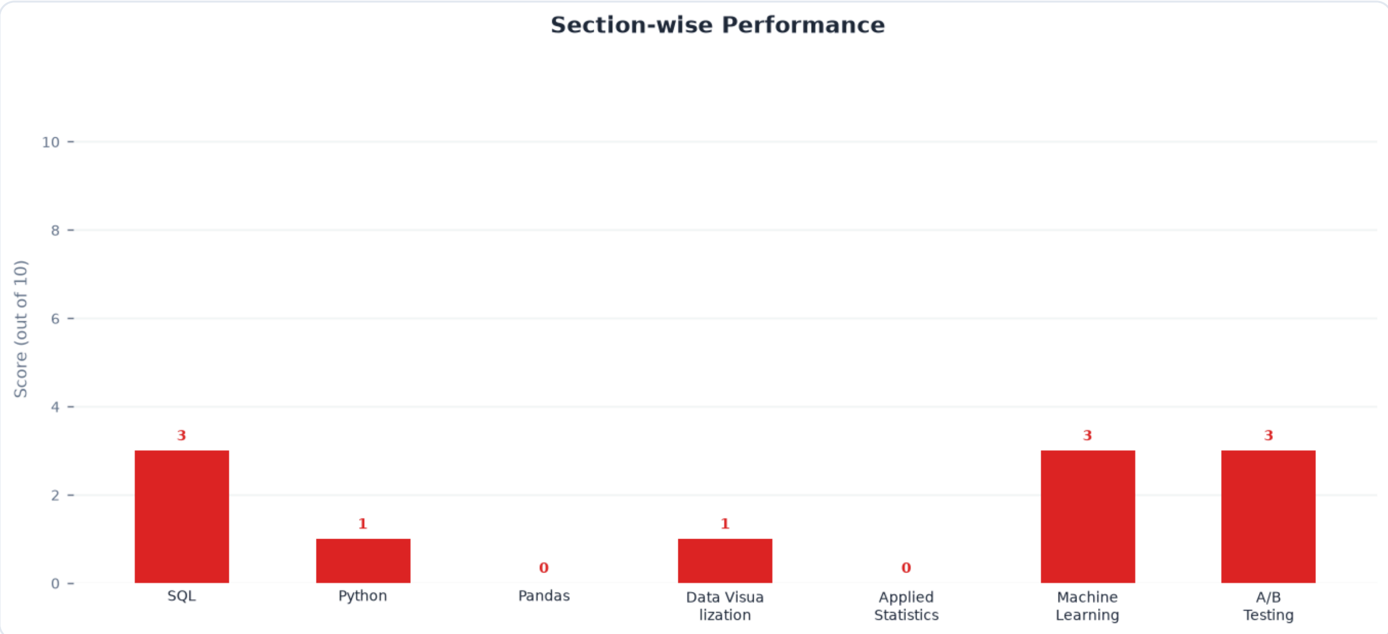
AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE	FINAL SCORE	CORRECT ANSWERS
Bob bb	**23** / 70 weighted points	**11** / 35 questions NEEDS IMPROVEMENT

Assessment Date: July 25, 2026 **Report Generated:** July 25, 2026 at 21:08 UTC **Role:** Data Science Expert

OVERALL RATING	SUCCESS RATE	WEIGHTED SCORE	TOP SKILL	NEEDS IMMEDIATE ATTENTION
★★☆☆☆	**32.9%**	**23 / 70**	**SQL**	**Pandas**



Skill Rating & Benchmark

Section	Your Score	Reference Benchmark (Demo)	Delta	Status
SQL	3 / 10	6.8	-3.8	Needs Work
Python	1 / 10	6.2	-5.2	Critical Gap
Pandas	0 / 10	5.8	-5.8	Critical Gap
Data Visualization	1 / 10	6.0	-5.0	Critical Gap
Applied Statistics	0 / 10	5.5	-5.5	Critical Gap
Machine Learning	3 / 10	5.2	-2.2	Needs Work
A/B Testing	3 / 10	6.4	-3.4	Needs Work

Attempt Statistics Breakdown

Section	Total Q	Attempted	Correct	Incorrect	Skipped	Accuracy
SQL	5	5	3	2	0	60%
Python	5	5	1	4	0	20%
Pandas	5	5	0	5	0	0%
Data Visualization	5	5	1	4	0	20%
Applied Statistics	5	5	0	5	0	0%
Machine Learning	5	5	3	2	0	60%
A/B Testing	5	5	3	2	0	60%

Executive Summary

The candidate answered 11 out of 35 questions correctly (31.4% accuracy), achieving a weighted score of 23 out of 70. This performance reveals critical gaps in core data science competencies, particularly in Python/Pandas execution details, statistical theory, and advanced experimental design. Targeted upskilling is required to meet the technical expectations of a professional data science role.

Key Strengths

- **Foundational SQL Querying:** Understands basic data retrieval and sorting operations.
- **Core Machine Learning Concepts:** Demonstrates a high-level conceptual awareness of ensemble models and feature importance.
- **Basic Visualization Awareness:** Recognizes standard plotting techniques and the necessity of addressing overplotting

Top Blind Spots

- **Pandas Execution & Memory Mechanics:** Gaps exist regarding index slicing behavior, DataFrame reshaping (`melt`, `concat`), and the modern Copy-on-Write (CoW) memory model.
- **Applied Probability & Statistics:** The candidate struggles with fundamental calculations, including the Central Limit Theorem, Standard Error, Bayes' Theorem, and sequential Beta-Binomial updating.
- **Advanced A/B Testing:** There is a lack of familiarity with experimental design challenges, specifically network interference (SUTVA violations) and variance reduction techniques like CUPED.

Actionable Learning Resources

Pandas & Python: Read the official Pandas documentation on "Copy-on-Write (CoW)" and complete the "Reshaping and Pivot Tables" user guide tutorials.

✓ Action Item

Probability & Statistics: Work through Chapter 2 (Bayes' Rule) and Chapter 3 (Common Distributions) of *Doing Bayesian Data Analysis* by John Kruschke.

✓ Action Item

A/B Testing: Study Chapter 10 (Interference) and Chapter 16 (Variance Reduction/CUPED) in *Trustworthy Online Controlled Experiments* by Kohavi, Tang, and Xu.

✓ Action Item

SQL Window Functions: Practice advanced window frame specifications (`RANGE` vs `ROWS`) using the interactive exercises on the PostgreSQLExercises platform.

✓ Action Item